

AI Engineer — GenAI, RAG & Agentic Systems (Graduate)

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PROFESSIONAL SUMMARY

AI Engineer building production-oriented Generative AI applications with Python, LangChain, LangGraph, RAG, and OpenAI-compatible LLM APIs (OpenAI, Groq). Shipped end-to-end AI systems — document Q&A, explainable ML triage, and agentic speech pipelines — as Dockerized, cloud-native FastAPI services with CI/CD automation and LangSmith observability. Eager to contribute within Agile/Scrum teams on cloud-native GenAI solutions from ideation to MVP.

TECHNICAL SKILLS

Languages: Python, SQL

GenAI & LLMs: Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Prompt Engineering, Structured Outputs (Pydantic), OpenAI API, Groq API, Llama 3.3 70B, LangSmith (Tracing & Observability)

Retrieval & Vector Search: ChromaDB, Sentence Transformers, Embeddings, Semantic Search, Document Chunking, Top-K Retrieval, PyMuPDF, Document Ingestion Pipelines

Backend & APIs: FastAPI, REST API Design, Uvicorn, Health/Readiness Endpoints, Swagger/OpenAPI, SQLite

Machine Learning & NLP: Scikit-learn, XGBoost, Pandas, NumPy, Model Evaluation (ROC-AUC, F1), Data Preprocessing, Feature Engineering, spaCy, Tokenization, Intent Recognition, Multilingual NLP, Whisper (Speech-to-Text)

Cloud, DevOps & Testing: Docker, Microservices, Containerization, CI/CD (GitHub Actions), Cloud Deployment (Hugging Face Spaces, Railway, Netlify), Git (Feature-Branch Workflow, Pull Requests), Pytest, Automated Testing, Hallucination Mitigation, Explainable AI

Methodologies: Ideation-to-MVP Delivery, Technical Documentation, eager to work within Agile/Scrum environments

EXPERIENCE

Web Developer Intern — iWebwala

Jan 2026 – Mar 2026

- Championed full-cycle development and launch of a solar company's website with domain registration and Netlify cloud deployment, establishing online presence from zero.
- Launched a conversational AI pipeline on Railway integrating Twilio Voice, Whisper, and ElevenLabs, automating 50% of initial customer inquiries.

PROJECTS

RAG Document Q&A System — FastAPI, LangChain, ChromaDB, Sentence Transformers, Groq (Llama-3.3-70B), LangSmith, Streamlit, Docker | *Live on Hugging Face Spaces*

- Architected an end-to-end RAG pipeline (PDF extraction, chunking, embeddings, ChromaDB retrieval, grounded LLM answers) in a Dockerized FastAPI backend, achieving 25 ms average retrieval latency and 100% retrieval hit rate across a 30-query benchmark.
- Implemented a cosine-similarity threshold returning "not found in document" instead of fabricated answers, and full LangSmith tracing with SQLite query logging (1047 ms avg / 3087 ms P95 across 50+ queries), fixing a retrieval-step bottleneck.

ExplainOps — AI Incident Triage Platform (2-Person Hackathon) — FastAPI, Scikit-learn, LangChain, Groq, SQLite, GitHub Actions | github.com/dhruv-motaval/ExplainOps

- Engineered the ML scoring engine and GenAI explanation module for a 2-person hackathon PoC fusing ticket, log, and chat event streams into ML-ranked incidents with plain-language LLM explanations, achieving 0.91 F1 and 0.90 ROC-AUC at 1.2 ms average inference latency.
- Built an explainable urgency-scoring pipeline with per-driver score attribution and deterministic fallback, ensuring auditable, hallucination-free triage even on LLM failure; collaborated via feature-branch Git workflow with pull requests.
- Implemented a CI/CD pipeline with GitHub Actions (Python 3.11/3.12 matrix) that retrains the model in-pipeline and runs end-to-end tests on every push, catching a missing-dependency bug on its first run.

Real-Time Multilingual Call Resolution System — LangGraph, FastAPI, Whisper, spaCy, Groq, Streamlit | github.com/dhruv-motaval/Real-Time-Multilingual-Call-Resolution-MyOnSite

- Engineered a 4-node LangGraph state machine (transcribe → language detection → intent recognition → deterministic routing) processing calls in English, Spanish, and Hindi with 100% local transcription and NLP.
- Implemented an event-sourced audit trail with state snapshots, AI-generated routing explanations, and event replay, delivering 100% decision traceability across every call.

EDUCATION

B.Tech, Artificial Intelligence — Parul Institute of Technology, Vadodara 2023 – 2026 | CGPA: 7.12 / 10

Diploma, Computer Engineering — A. Y. Dadabhai Institute of Technology, Kosamba 2019 – 2023 | CGPA: 7.91 / 10

Publication: "Limits of AI in Software Development: Empirical Analysis of AI vs Human Engineers," ISJEM, April 2026 (DOI: 10.55041/ISJEM06513)